

Factor Analysis

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.910
Bartlett's Test of Sphericity	Approx. Chi-Square	25169.787
	df	378
	Sig.	.000

Communalities

	Initial	Extraction
TDQ1	.573	.670
TDQ2	.576	.668
TDQ3	.529	.611
TDQ4	.575	.665
TC1	.600	.696
TC2	.552	.631
TC3	.596	.683
TC4	.570	.650
TCR1	.609	.712
TCR2	.557	.640
TCR3	.574	.655
TCR4	.534	.611
TSR1	.576	.658
TSR2	.606	.700
TSR3	.565	.643
TSR4	.582	.669
SA1	.631	.710
SA2	.607	.669
SA3	.623	.700
SA4	.648	.734
EMP1	.575	.657
EMP2	.605	.695
EMP3	.578	.661
EMP4	.578	.661
PTEO1	.609	.697
PTEO2	.586	.660
PTEO3	.615	.700
PTEO4	.589	.672

Extraction Method: Principal Axis
Factoring.

Total Variance Explained

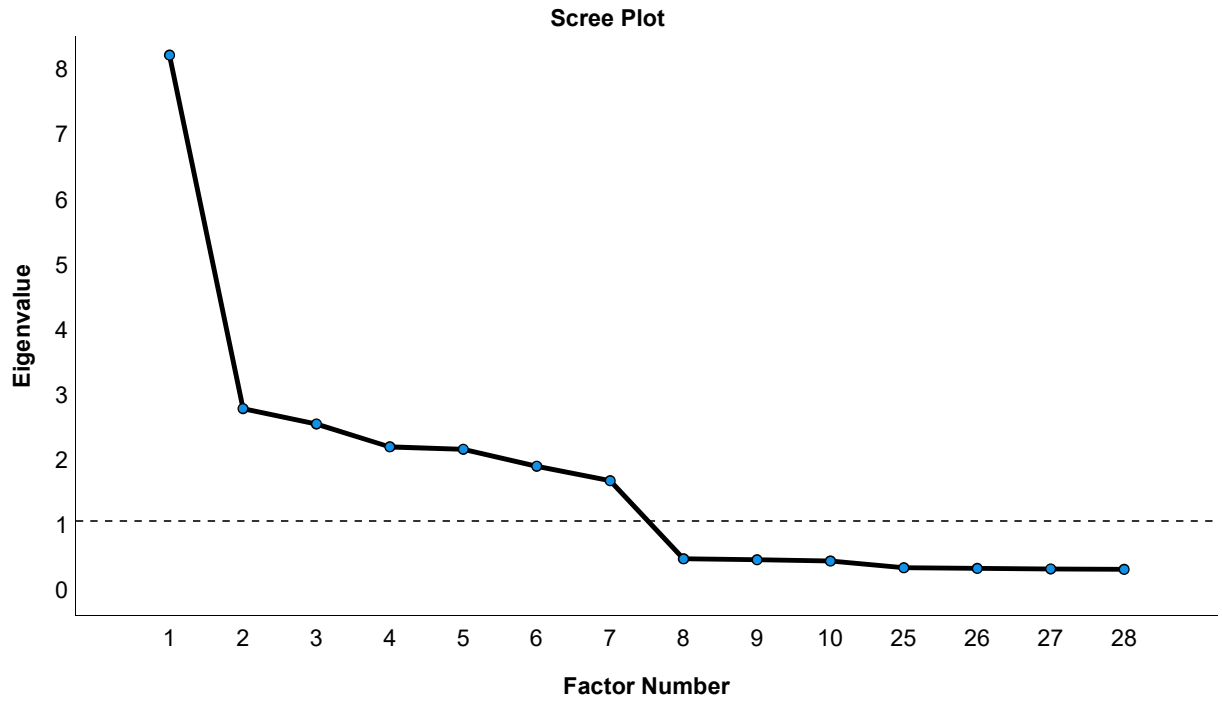
Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings	
	Total	% of Variance	Cumulative %	Total	% of Variance
1	8.159	29.138	29.138	7.833	27.976
2	2.727	9.738	38.876	2.397	8.560
3	2.492	8.898	47.774	2.151	7.682
4	2.139	7.640	55.414	1.797	6.417
5	2.103	7.509	62.924	1.770	6.323
6	1.842	6.579	69.503	1.536	5.487
7	1.620	5.785	75.288	1.293	4.620
8	.419	1.496	76.784		
9	.404	1.445	78.229		
10	.386	1.378	79.607		
11	.377	1.346	80.953		
12	.372	1.329	82.282		
13	.361	1.291	83.572		
14	.358	1.278	84.851		
15	.350	1.250	86.101		
16	.346	1.235	87.336		
17	.338	1.207	88.543		
18	.329	1.175	89.717		
19	.317	1.132	90.849		
20	.312	1.116	91.964		
21	.302	1.080	93.045		
22	.295	1.052	94.096		
23	.293	1.046	95.143		
24	.284	1.014	96.157		
25	.281	1.005	97.162		
26	.272	.972	98.135		
27	.265	.945	99.080		
28	.258	.920	100.000		

Total Variance Explained

Factor	Extraction Sums of Squared ...	Rotation Sums of Squared Loadings ^a
	Cumulative %	Total
1	27.976	5.046
2	36.536	4.561
3	44.218	4.054
4	50.635	4.013
5	56.958	4.771
6	62.445	3.870
7	67.064	4.150
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		
25		
26		
27		
28		

Extraction Method: Principal Axis Factoring.

a. When factors are correlated, sums of squared loadings cannot be added to obtain a total variance.



Factor Matrix^a

	Factor					
	1	2	3	4	5	6
TDQ1			.523			
TDQ2			.509			
TDQ3			.527			
TDQ4			.535			
TC1						
TC2						
TC3	.507					
TC4						
TCR1	.522			-.518		
TCR2	.503					
TCR3	.517					
TCR4						
TSR1						
TSR2	.505					
TSR3						
TSR4						

Factor Matrix^a

	Factor 7
TDQ1	
TDQ2	
TDQ3	
TDQ4	
TC1	
TC2	
TC3	
TC4	
TCR1	
TCR2	
TCR3	
TCR4	
TSR1	
TSR2	
TSR3	
TSR4	

Factor Matrix^a

	Factor					
	1	2	3	4	5	6
SA1	.628					-.528
SA2	.624					
SA3	.617					-.525
SA4	.638					-.525
EMP1	.562					
EMP2	.589					
EMP3	.578					
EMP4	.582					
PTEO1	.550					
PTEO2	.552					
PTEO3	.572					
PTEO4	.541					

Factor Matrix^a

	Factor 7
SA1	
SA2	
SA3	
SA4	
EMP1	
EMP2	
EMP3	
EMP4	
PTEO1	
PTEO2	
PTEO3	
PTEO4	

Extraction Method: Principal Axis Factoring.

a. 7 factors extracted. 6 iterations required.

Pattern Matrix^a

	Factor					
	1	2	3	4	5	6
TDQ1						.822
TDQ2						.807
TDQ3						.789
TDQ4						.811
TC1				.836		
TC2				.806		
TC3				.814		
TC4				.797		
TCR1						
TCR2						
TCR3						
TCR4						
TSR1			.808			
TSR2			.832			
TSR3			.805			

Pattern Matrix^a

	Factor 7
TDQ1	
TDQ2	
TDQ3	
TDQ4	
TC1	
TC2	
TC3	
TC4	
TCR1	.844
TCR2	.794
TCR3	.795
TCR4	.793
TSR1	
TSR2	
TSR3	

Pattern Matrix^a

	Factor					
	1	2	3	4	5	6
TSR4			.817			
SA1	.843					
SA2	.799					
SA3	.843					
SA4	.855					
EMP1					.807	
EMP2					.828	
EMP3					.812	
EMP4					.811	
PTEO1		.836				
PTEO2		.800				
PTEO3		.826				
PTEO4		.829				

Pattern Matrix^a

	Factor 7
TSR4	
SA1	
SA2	
SA3	
SA4	
EMP1	
EMP2	
EMP3	
EMP4	
PTEO1	
PTEO2	
PTEO3	
PTEO4	

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.^a
a. Rotation converged in 6 iterations.

Structure Matrix

	Factor					
	1	2	3	4	5	6
TDQ1						.817
TDQ2						.817
TDQ3						.780
TDQ4						.815
TC1				.834		
TC2				.793		
TC3				.826		
TC4				.805		
TCR1						
TCR2						
TCR3						
TCR4						
TSR1			.811			
TSR2			.836			
TSR3			.800			
TSR4			.817			
SA1	.843					
SA2	.817					
SA3	.836					
SA4	.856					
EMP1					.810	
EMP2					.833	
EMP3					.812	
EMP4					.813	
PTEO1		.834				
PTEO2		.811				
PTEO3		.835				
PTEO4		.819				

Structure Matrix

	Factor 7
TDQ1	
TDQ2	
TDQ3	
TDQ4	
TC1	
TC2	
TC3	
TC4	
TCR1	.843
TCR2	.799
TCR3	.808
TCR4	.780
TSR1	
TSR2	
TSR3	
TSR4	
SA1	
SA2	
SA3	
SA4	
EMP1	
EMP2	
EMP3	
EMP4	
PTEO1	
PTEO2	
PTEO3	
PTEO4	

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.

Factor Correlation Matrix

Factor	1	2	3	4	5	6
1	1.000	.400	.345	.316	.441	.344
2	.400	1.000	.272	.247	.494	.270
3	.345	.272	1.000	.341	.261	.225
4	.316	.247	.341	1.000	.291	.235
5	.441	.494	.261	.291	1.000	.262
6	.344	.270	.225	.235	.262	1.000
7	.373	.247	.287	.287	.323	.310

Factor Correlation Matrix

Factor	7
1	.373
2	.247
3	.287
4	.287
5	.323
6	.310
7	1.000

Extraction Method: Principal Axis Factoring.

Rotation Method: Promax with Kaiser Normalization.

Multilayer Perceptron

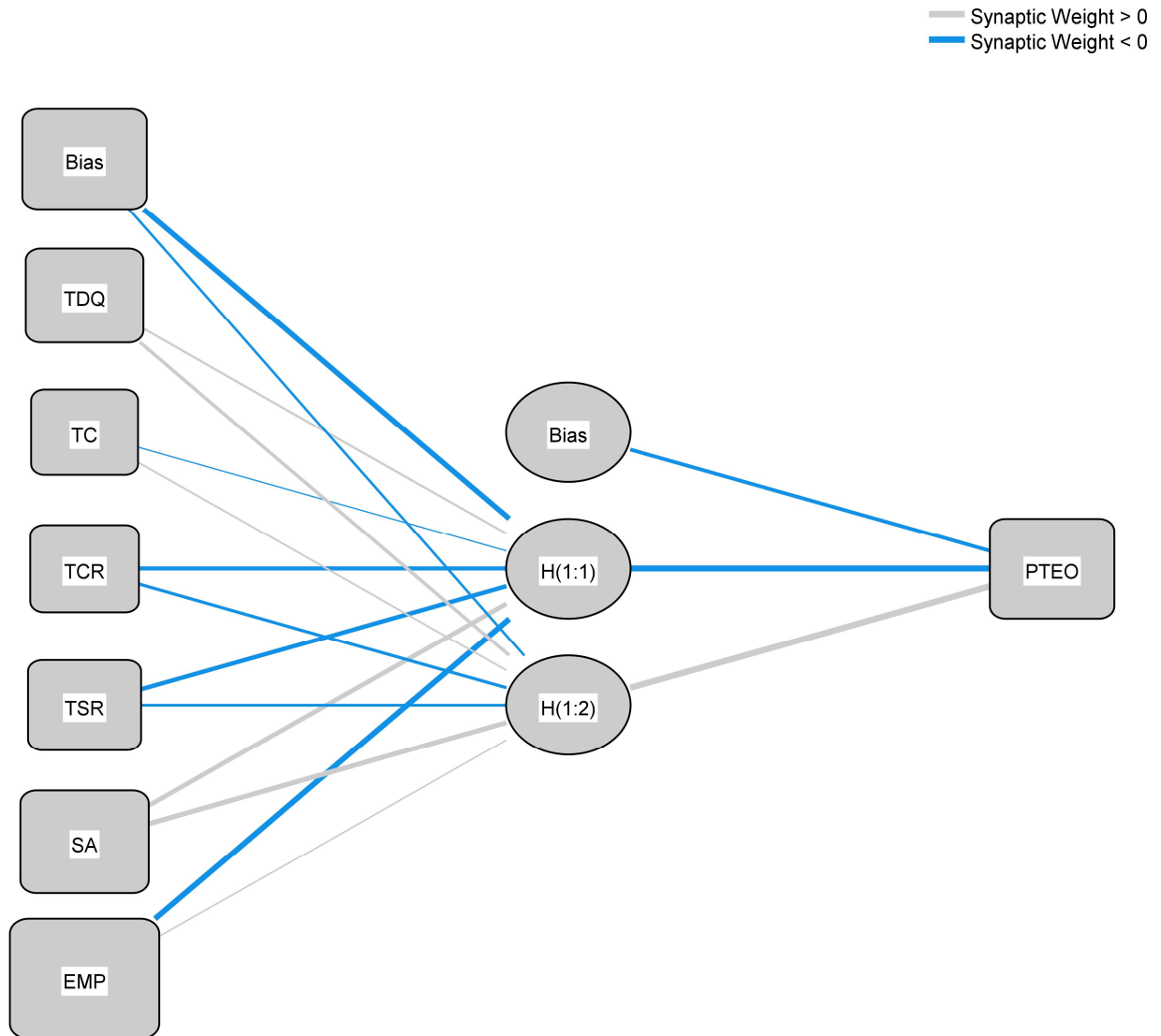
Case Processing Summary

	N	Percent
Sample		
Training	1046	69.7%
Testing	454	30.3%
Valid	1500	100.0%
Excluded	0	
Total	1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

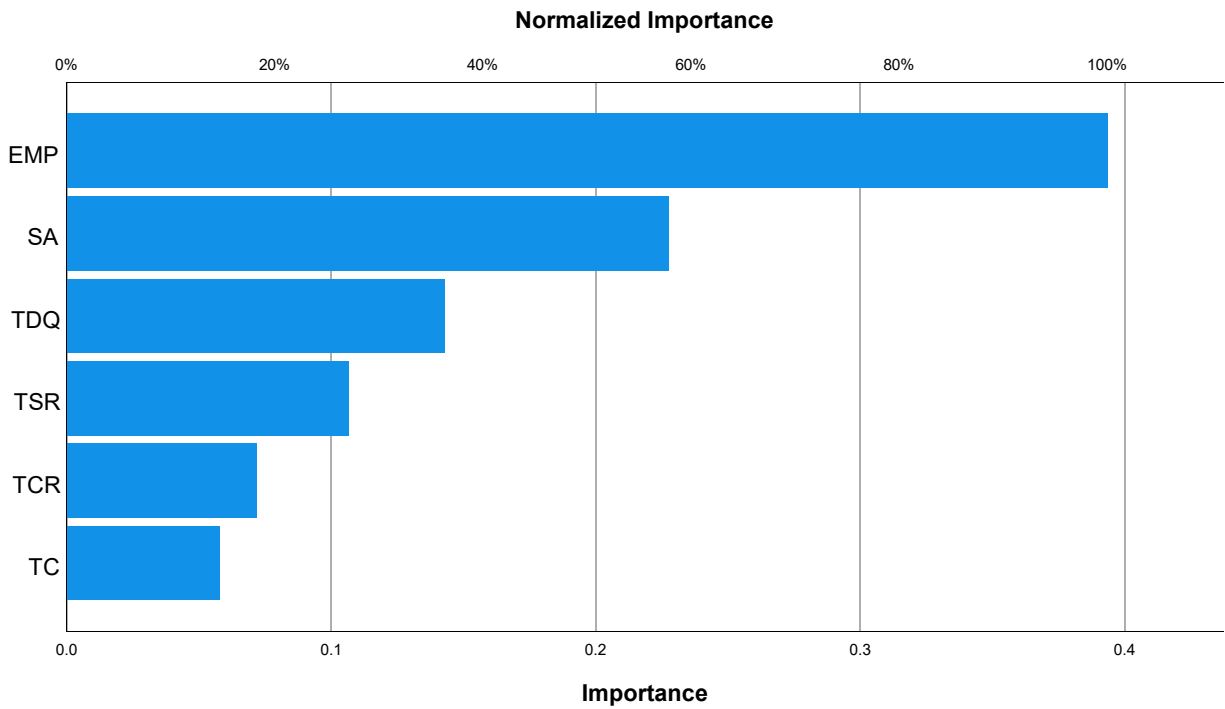
Training	Sum of Squares Error	386.972
	Relative Error	.741
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.09
Testing	Sum of Squares Error	155.784
	Relative Error	.754

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.143	36.3%
TC	.058	14.7%
TCR	.072	18.2%
TSR	.107	27.1%
SA	.228	57.8%
EMP	.393	100.0%



Multilayer Perceptron

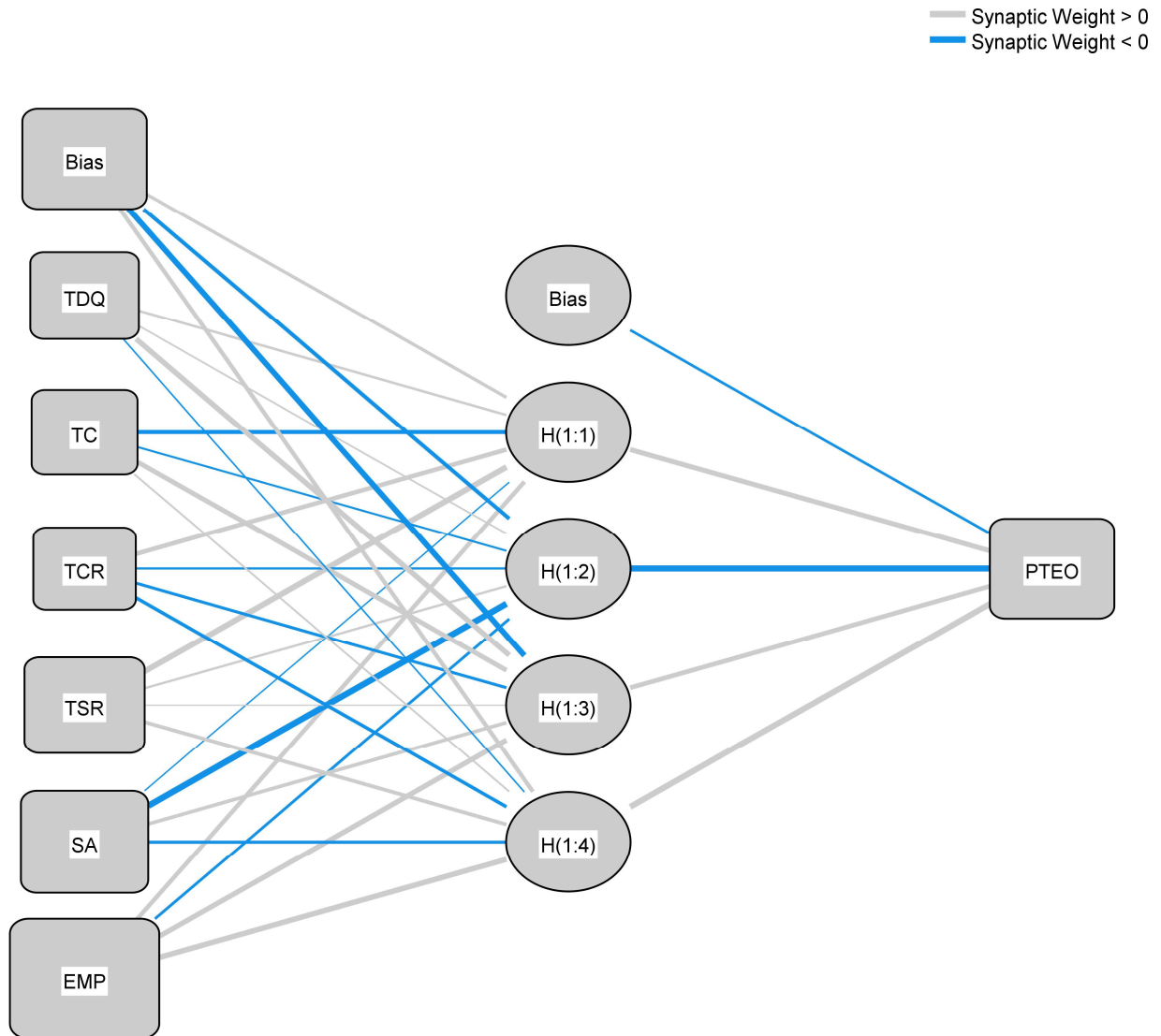
Case Processing Summary

		N	Percent
Sample	Training	1053	70.2%
	Testing	447	29.8%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

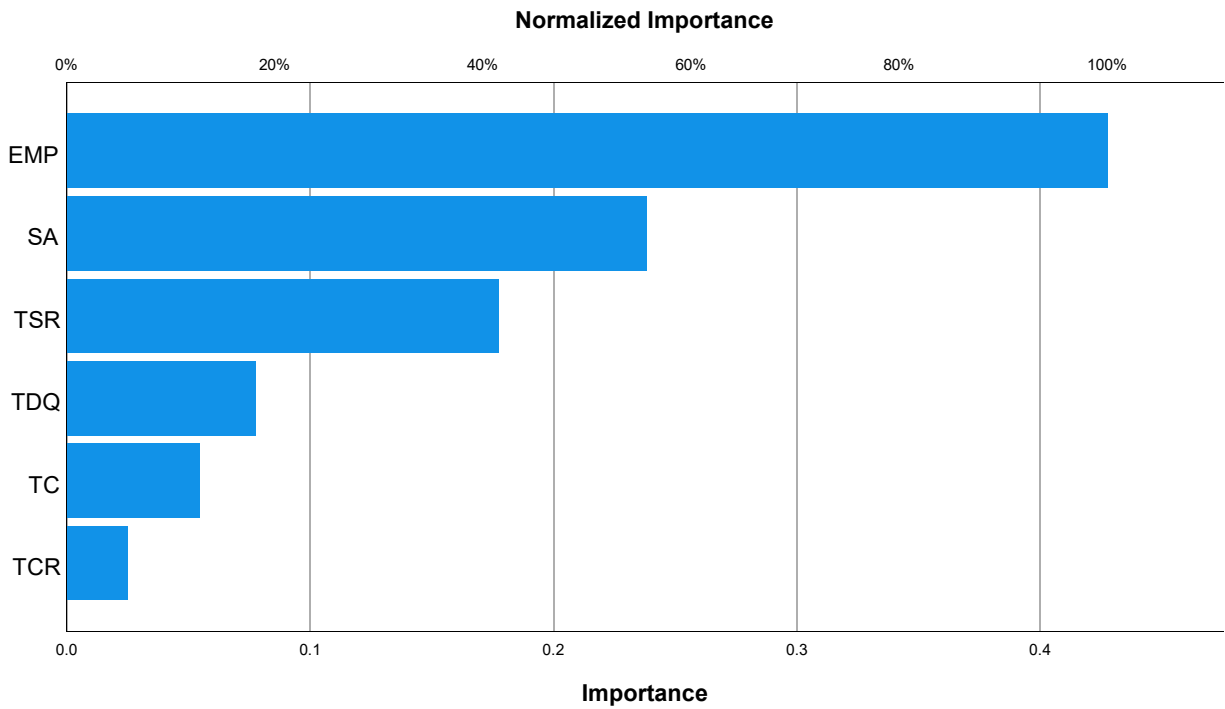
Training	Sum of Squares Error	396.936
	Relative Error	.755
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.09
Testing	Sum of Squares Error	156.290
	Relative Error	.727

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.078	18.1%
TC	.054	12.7%
TCR	.025	5.9%
TSR	.177	41.4%
SA	.238	55.7%
EMP	.428	100.0%



Multilayer Perceptron

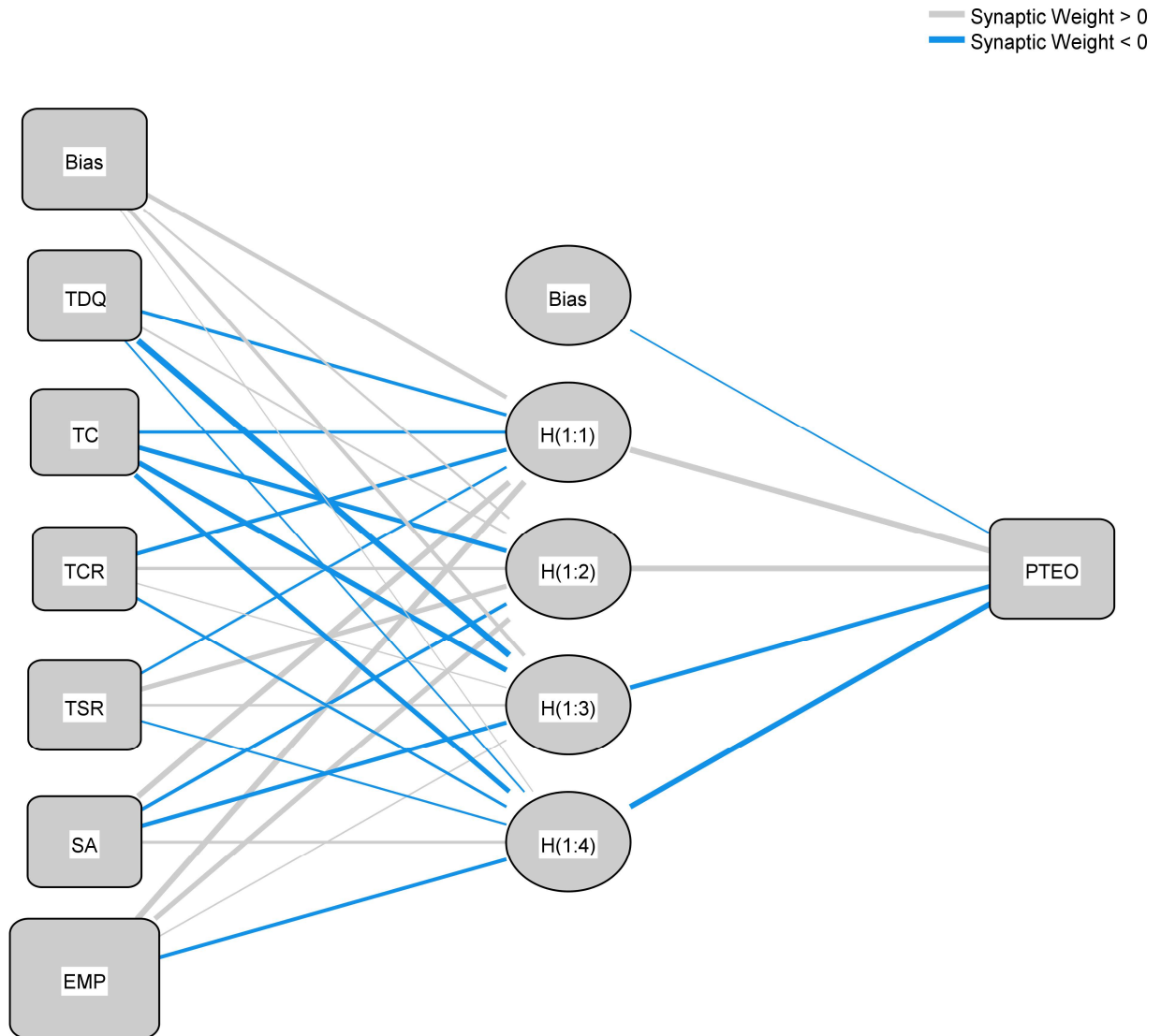
Case Processing Summary

		N	Percent
Sample	Training	1033	68.9%
	Testing	467	31.1%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

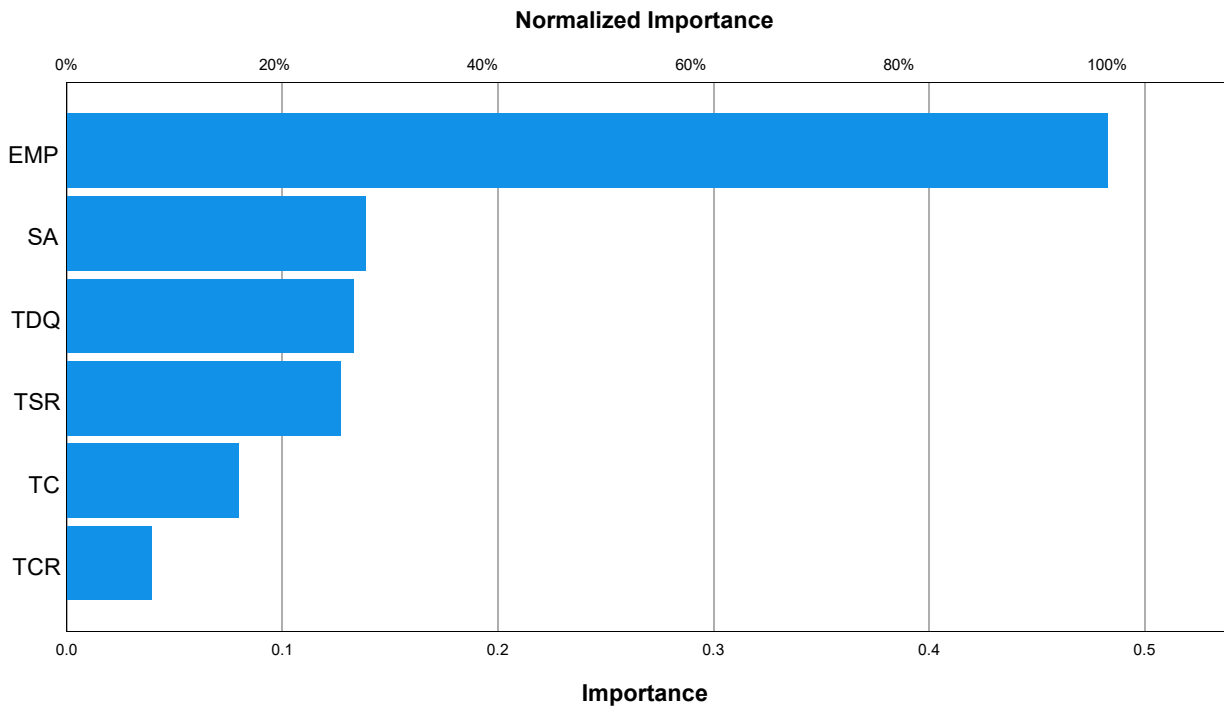
Training	Sum of Squares Error	386.189
	Relative Error	.748
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.09
Testing	Sum of Squares Error	176.330
	Relative Error	.738

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.133	27.5%
TC	.080	16.5%
TCR	.039	8.1%
TSR	.127	26.3%
SA	.139	28.7%
EMP	.483	100.0%



Multilayer Perceptron

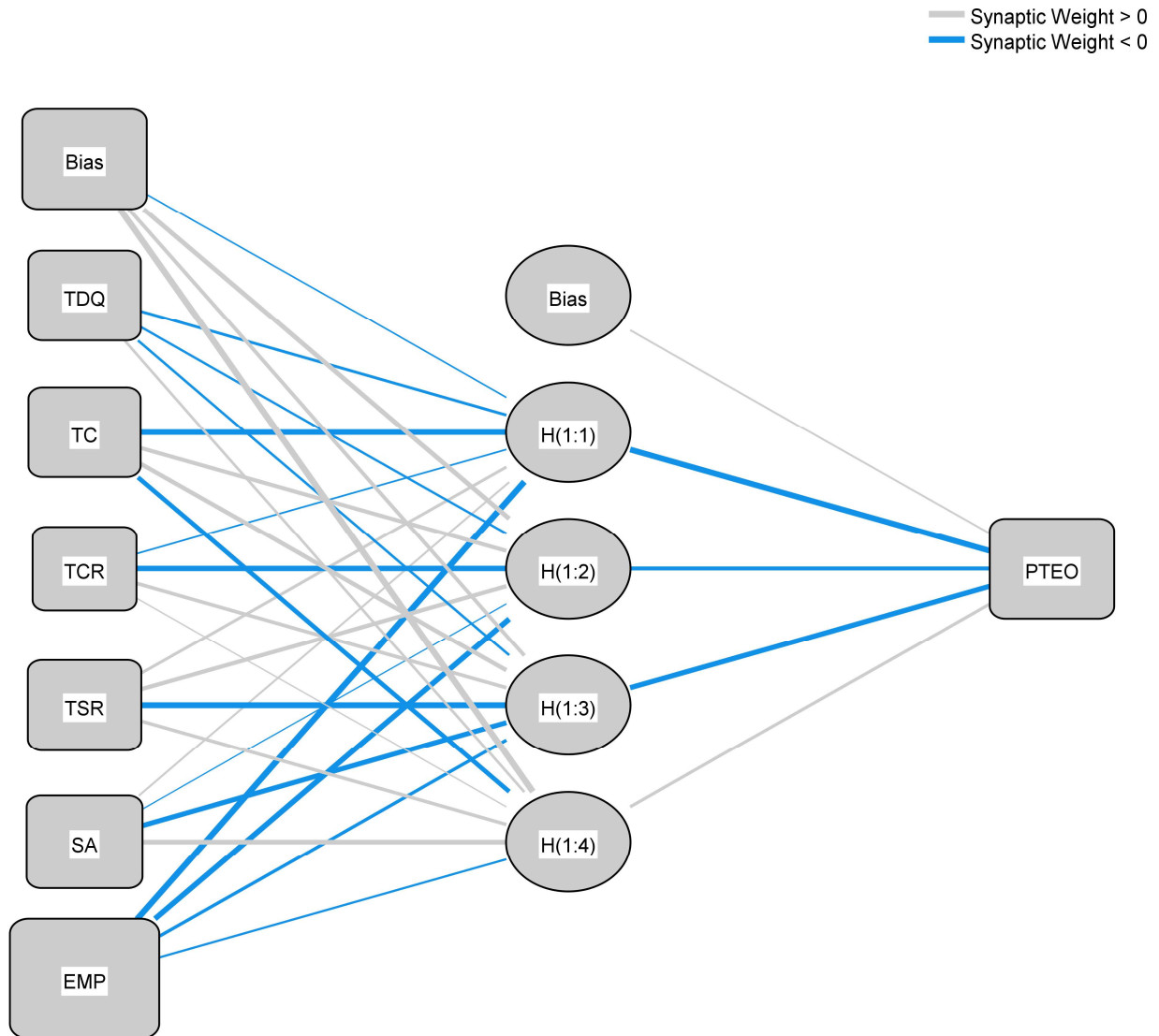
Case Processing Summary

		N	Percent
Sample	Training	1066	71.1%
	Testing	434	28.9%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

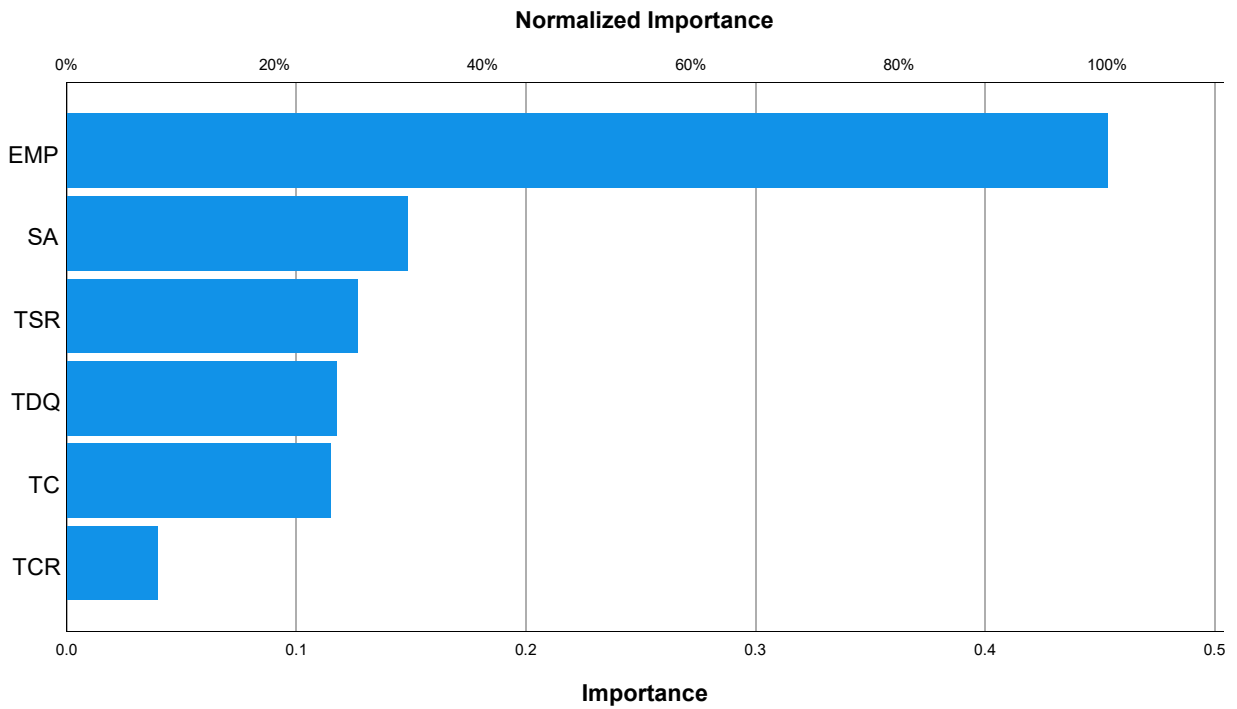
Training	Sum of Squares Error	402.451
	Relative Error	.756
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.08
Testing	Sum of Squares Error	159.113
	Relative Error	.740

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.118	25.9%
TC	.115	25.3%
TCR	.040	8.7%
TSR	.127	27.9%
SA	.148	32.7%
EMP	.453	100.0%



Multilayer Perceptron

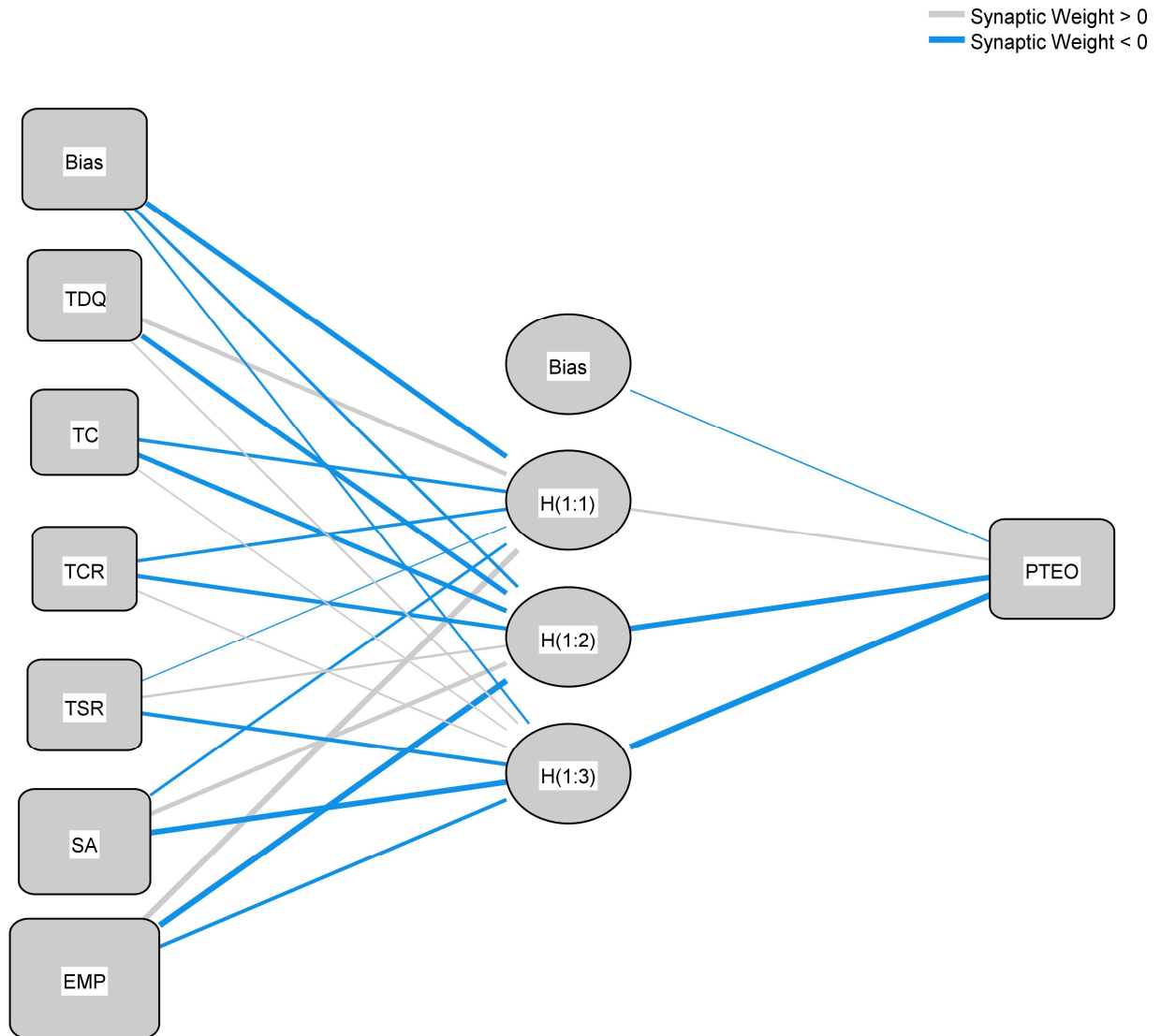
Case Processing Summary

		N	Percent
Sample	Training	1055	70.3%
	Testing	445	29.7%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

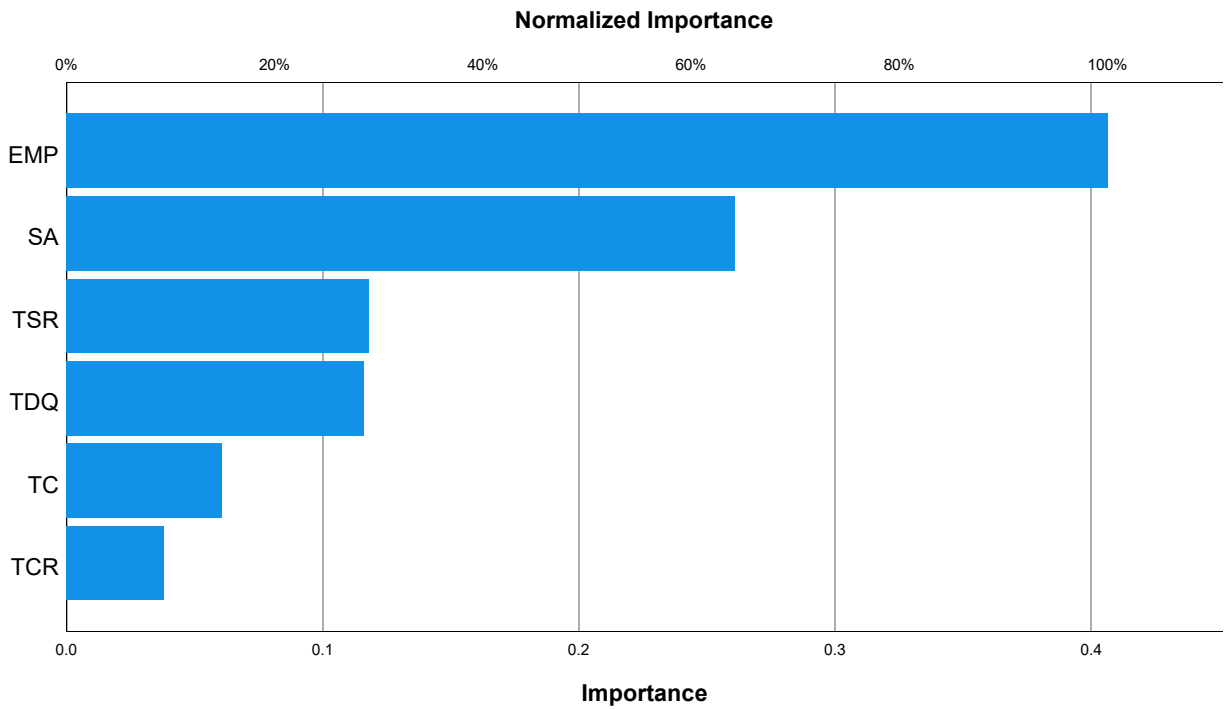
Training	Sum of Squares Error	386.239
	Relative Error	.733
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.09
Testing	Sum of Squares Error	165.982
	Relative Error	.785

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.116	28.5%
TC	.061	14.9%
TCR	.038	9.4%
TSR	.118	29.0%
SA	.261	64.2%
EMP	.406	100.0%



Multilayer Perceptron

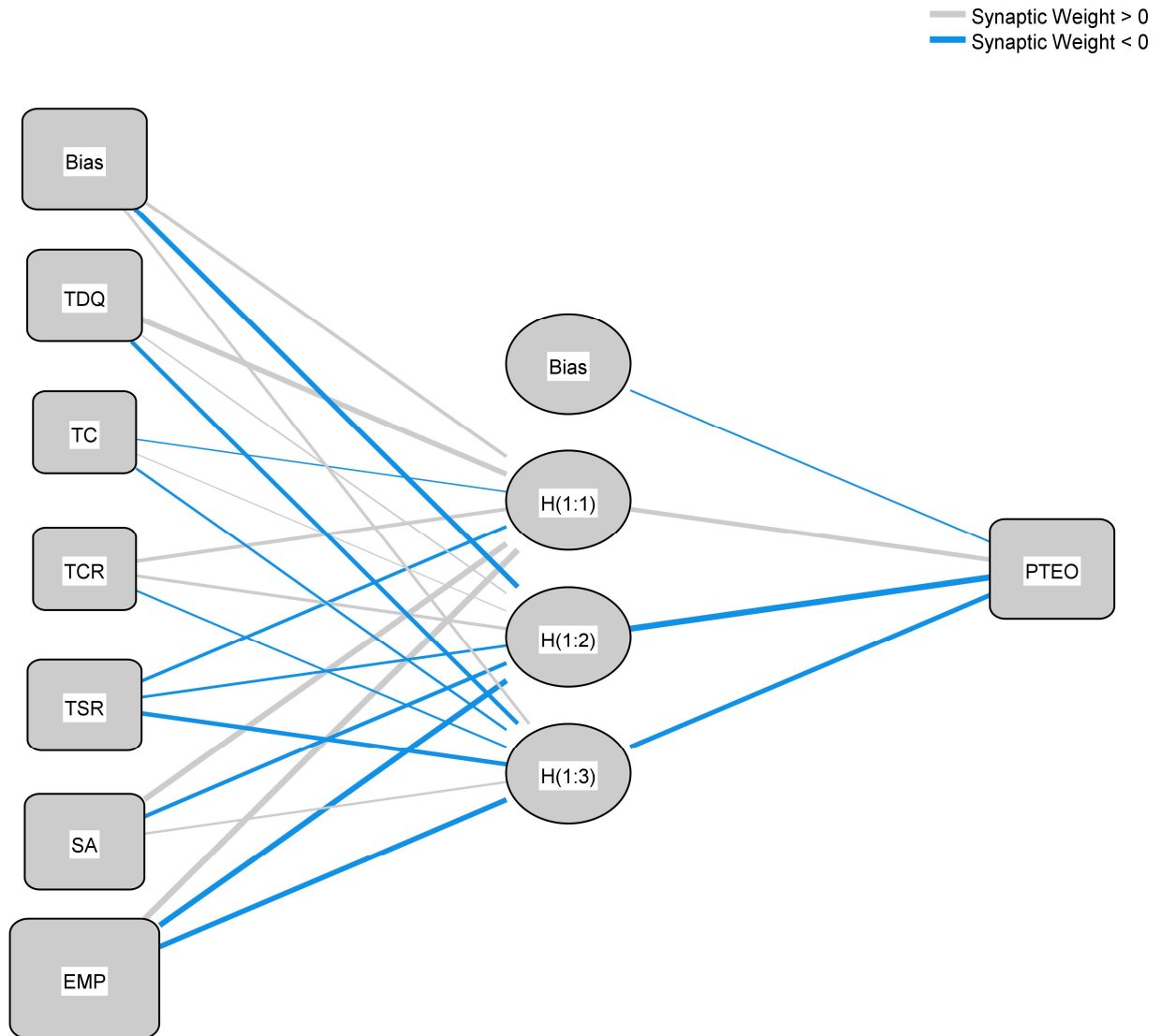
Case Processing Summary

		N	Percent
Sample	Training	1062	70.8%
	Testing	438	29.2%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

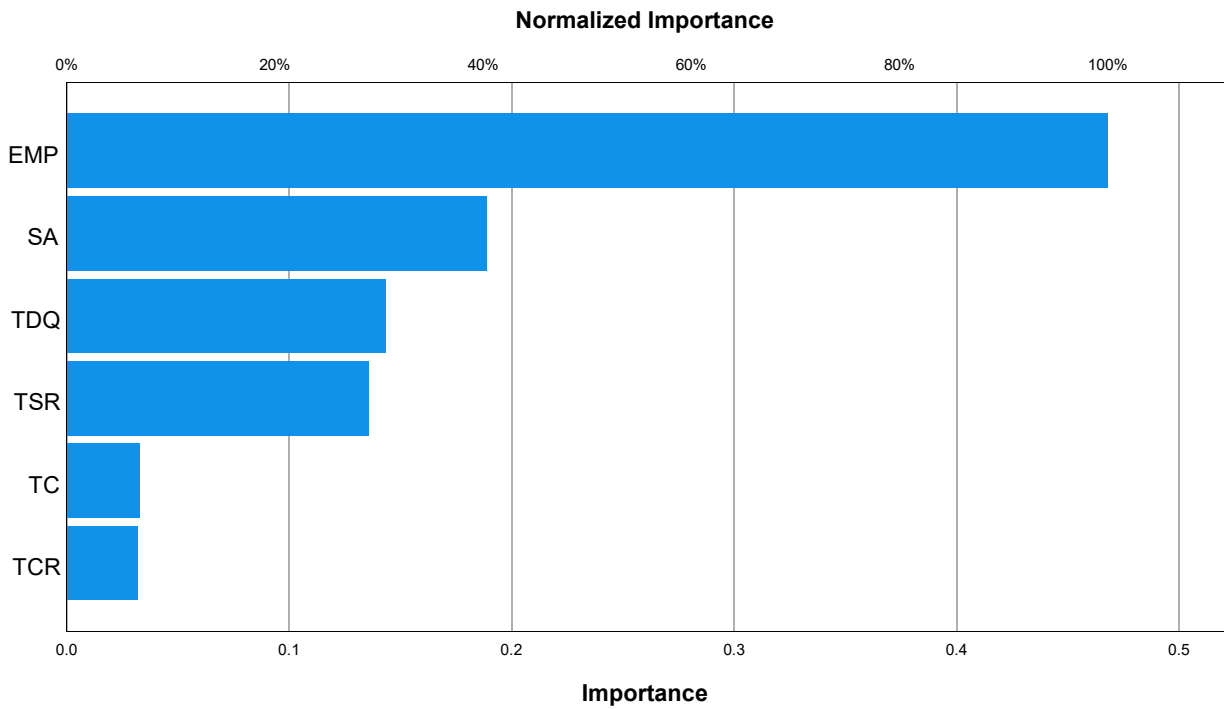
Training	Sum of Squares Error	387.928
	Relative Error	.731
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.09
Testing	Sum of Squares Error	177.035
	Relative Error	.812

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.143	30.6%
TC	.033	7.0%
TCR	.032	6.8%
TSR	.136	29.0%
SA	.189	40.3%
EMP	.468	100.0%



Multilayer Perceptron

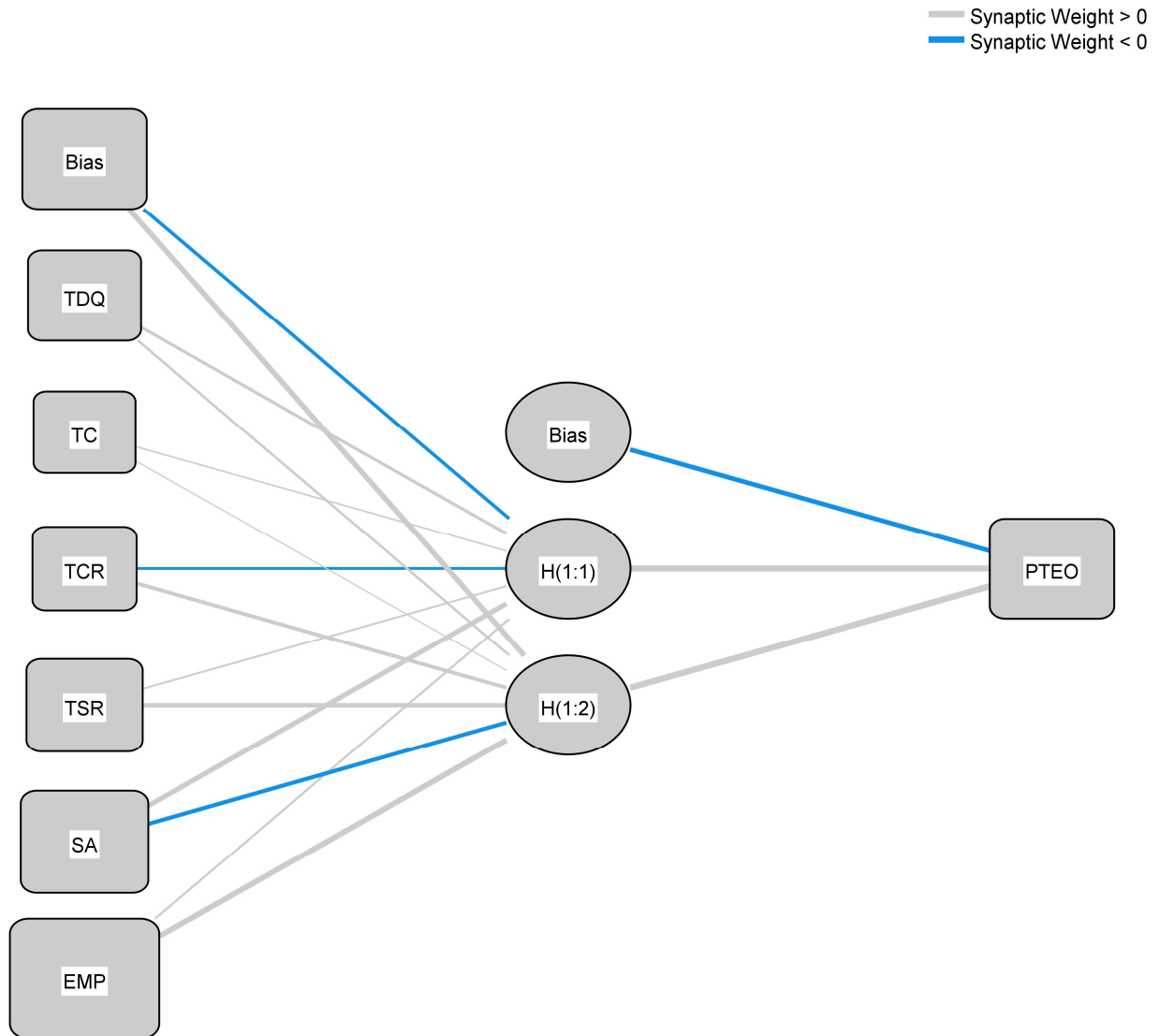
Case Processing Summary

		N	Percent
Sample	Training	1061	70.7%
	Testing	439	29.3%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

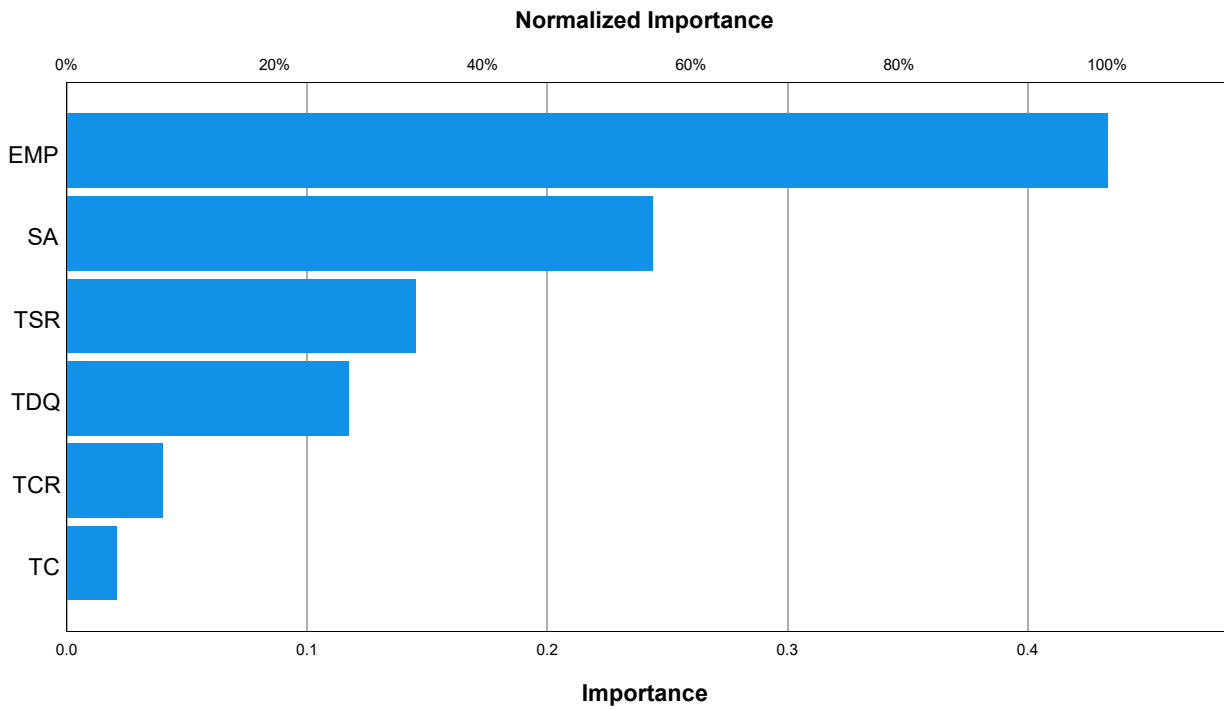
Training	Sum of Squares Error	391.665
	Relative Error	.739
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.12
Testing	Sum of Squares Error	166.095
	Relative Error	.750

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.117	27.1%
TC	.021	4.8%
TCR	.040	9.2%
TSR	.145	33.5%
SA	.244	56.3%
EMP	.433	100.0%



Multilayer Perceptron

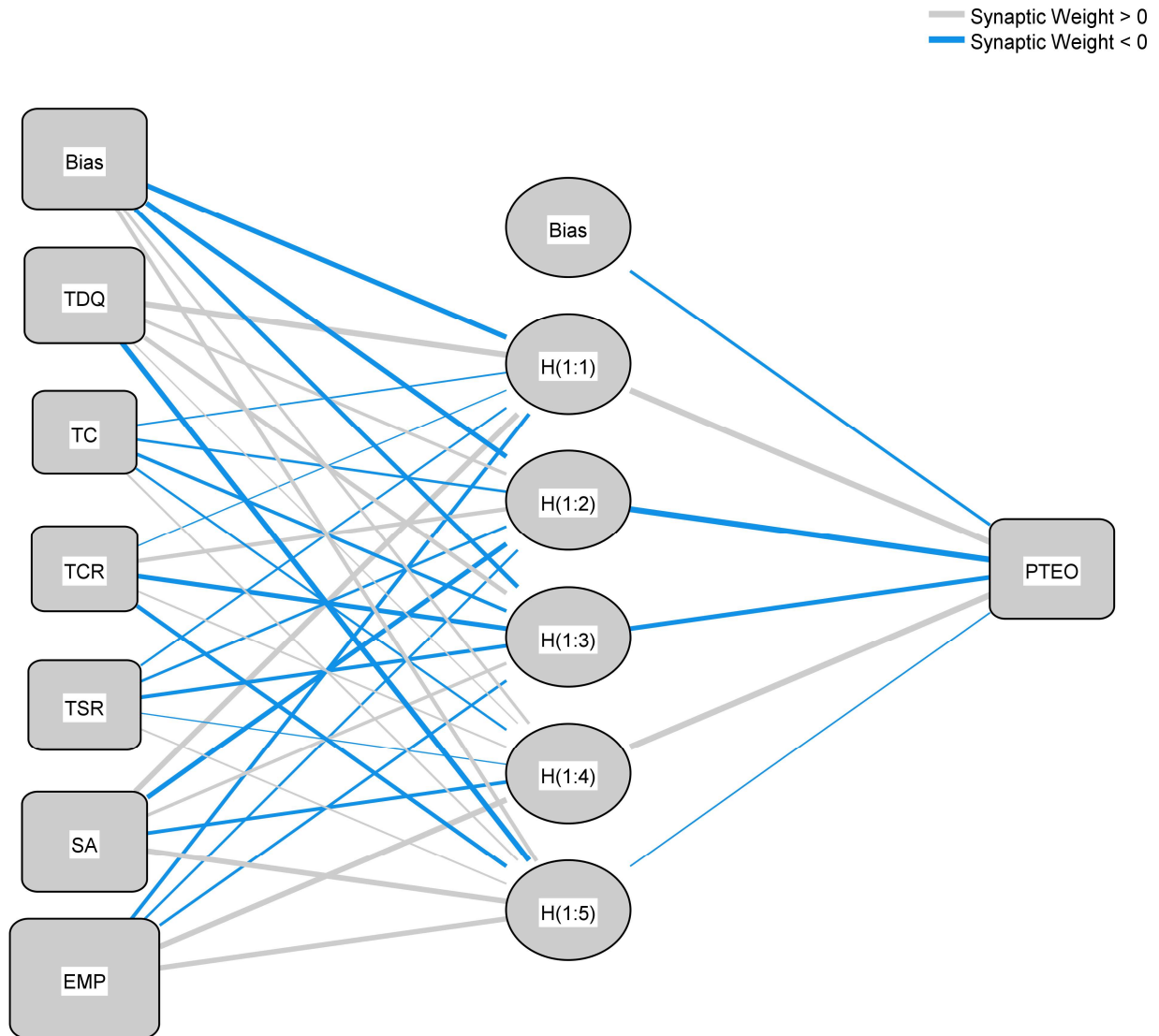
Case Processing Summary

		N	Percent
Sample	Training	1045	69.7%
	Testing	455	30.3%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		5
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

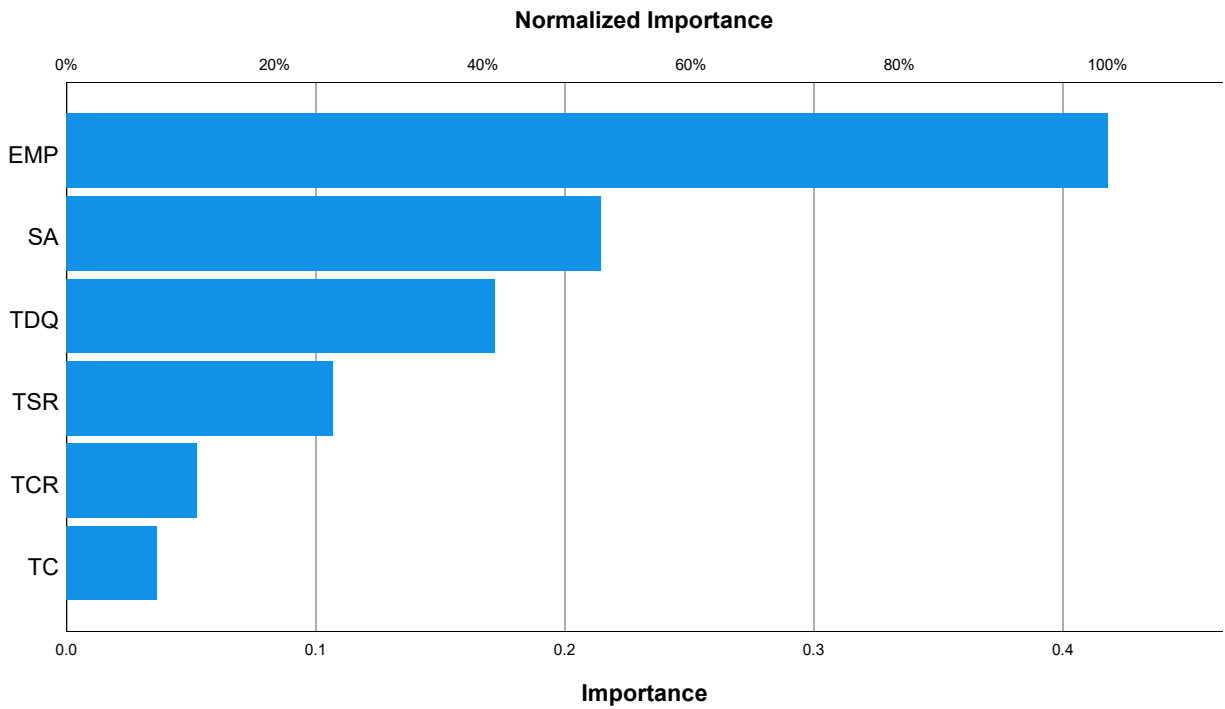
Training	Sum of Squares Error	376.555
	Relative Error	.721
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.22
Testing	Sum of Squares Error	166.871
	Relative Error	.781

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.172	41.1%
TC	.036	8.7%
TCR	.052	12.5%
TSR	.107	25.6%
SA	.214	51.3%
EMP	.418	100.0%



Multilayer Perceptron

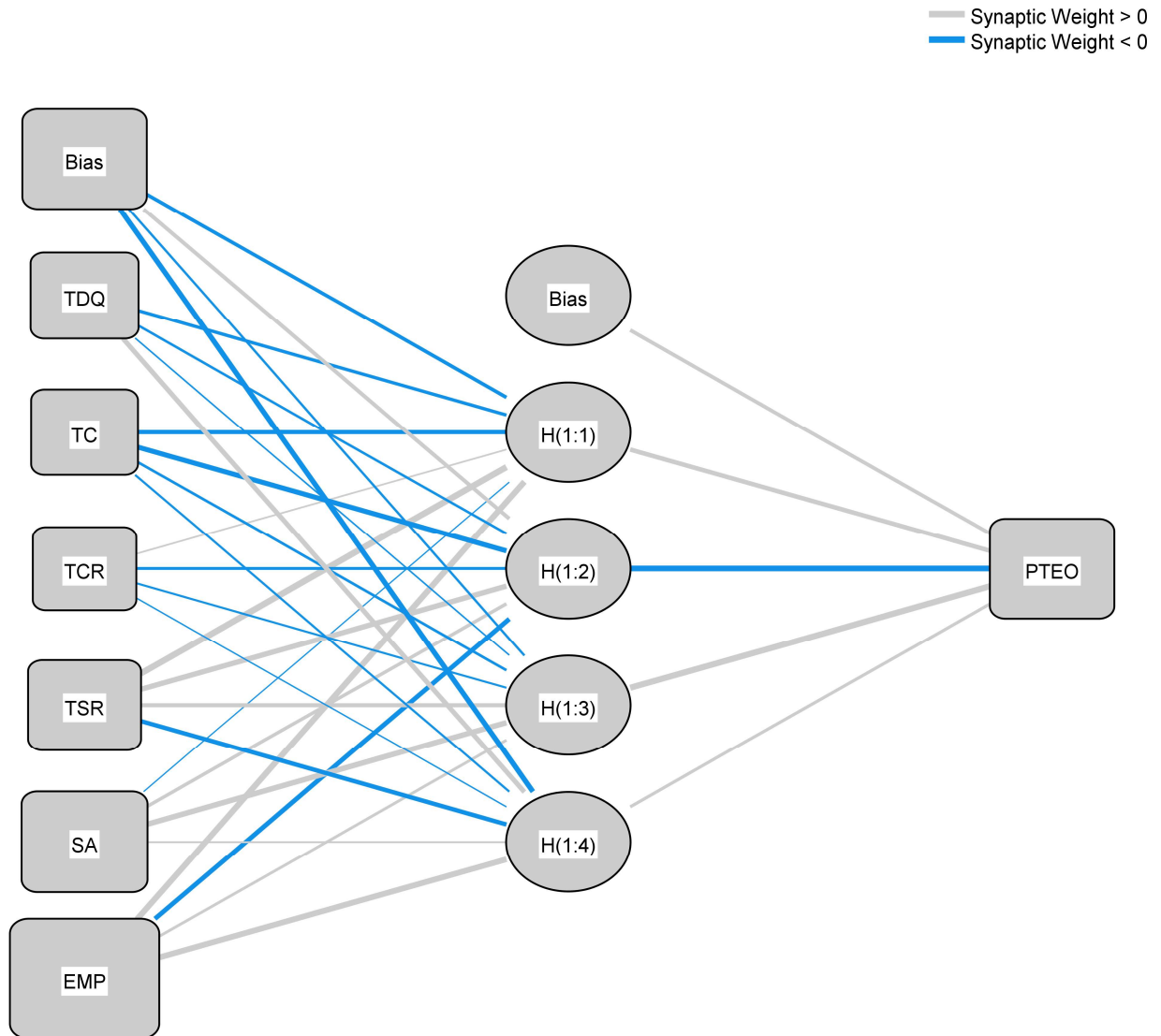
Case Processing Summary

		N	Percent
Sample	Training	1067	71.1%
	Testing	433	28.9%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

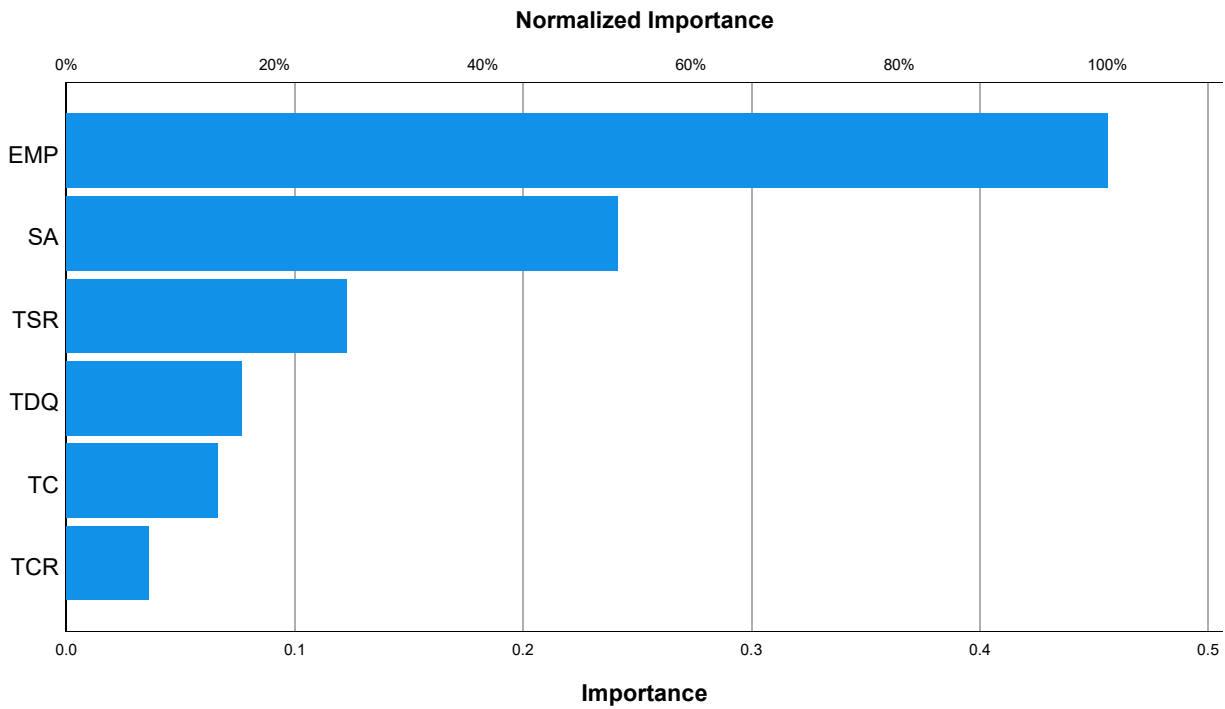
Training	Sum of Squares Error	397.465
	Relative Error	.746
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.08
Testing	Sum of Squares Error	154.288
	Relative Error	.753

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.077	16.9%
TC	.066	14.6%
TCR	.036	8.0%
TSR	.123	26.9%
SA	.242	53.0%
EMP	.456	100.0%



Multilayer Perceptron

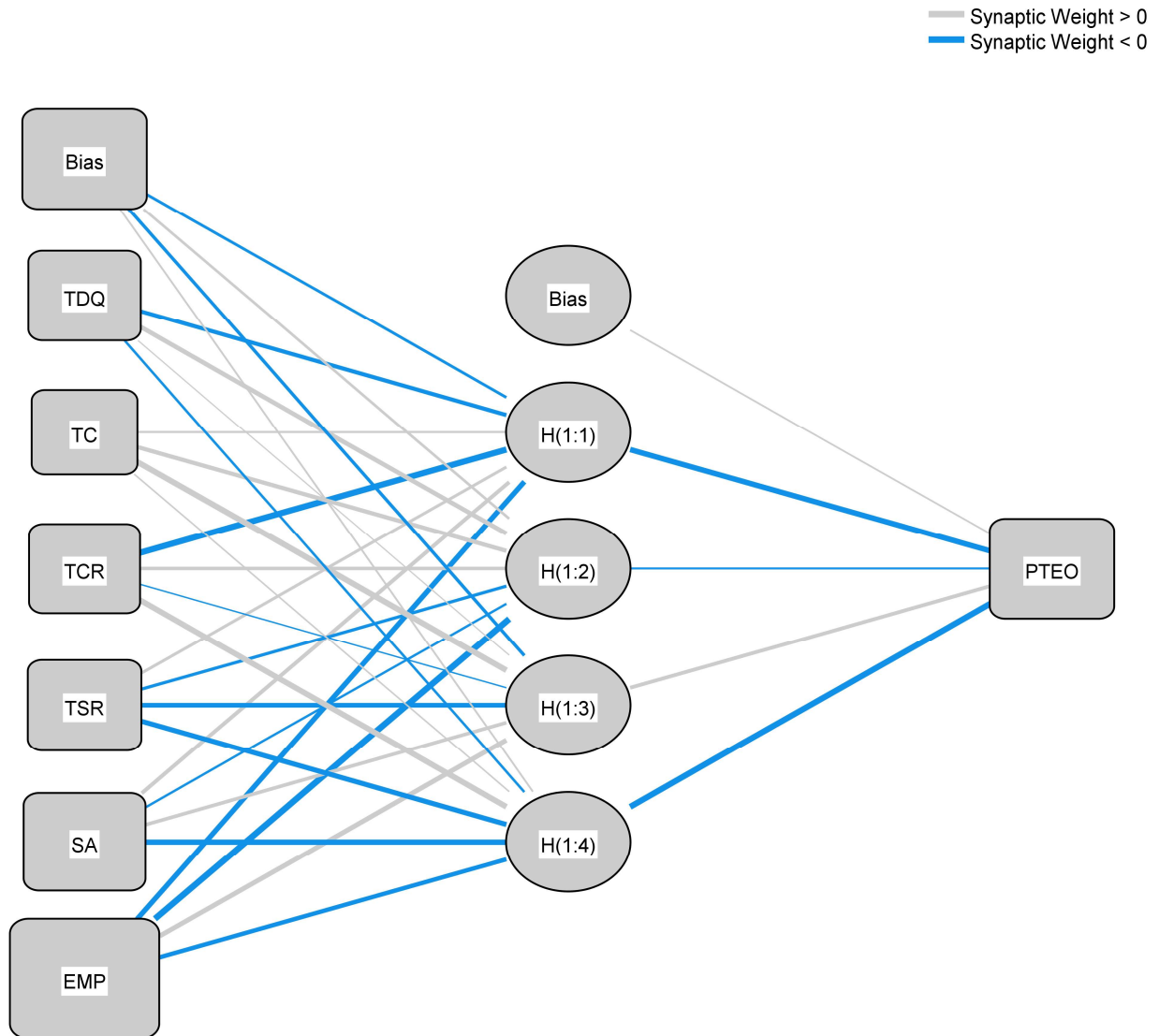
Case Processing Summary

		N	Percent
Sample	Training	1067	71.1%
	Testing	433	28.9%
Valid		1500	100.0%
Excluded		0	
Total		1500	

Network Information

Input Layer	Covariates	1	TDQ
		2	TC
		3	TCR
		4	TSR
		5	SA
		6	EMP
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	PTEO
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

Training	Sum of Squares Error	412.897
	Relative Error	.775
	Stopping Rule Used	1 consecutive step (s) with no decrease in error ^a
	Training Time	0:00:00.13
Testing	Sum of Squares Error	152.921
	Relative Error	.679

Dependent Variable: PTEO

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TDQ	.110	25.8%
TC	.053	12.5%
TCR	.098	23.0%
TSR	.120	28.2%
SA	.191	44.9%
EMP	.427	100.0%

